PT SAPTAINDRA SEJATI

DIFFERENTIAL

RECEIVING INSPECTION

CHECKSHEET

MACHINE MODEL

HD785-7

RO NO :

POWERTRAIN SECTION

PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

DOCUMENT NO QRC-09-PT-031

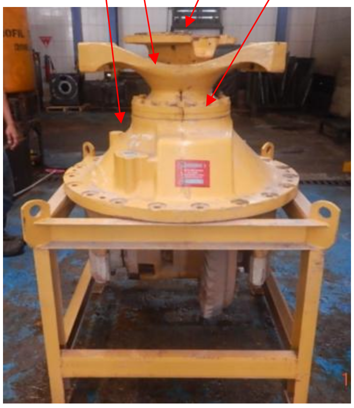
RECEIVING INSPECTION SHEET **DIFFERENTIAL**	MACHINE MODEL	HD785-7	PART NO		HOUR METER	
	COMP. MODEL		COMP. S/N		LIFE TIME	
	RECEIVING DATE		SITE		REVISION NO	
	HISTORYCAL		WO NO		PAGE	1

BELOW ARE THE ITEMS RELATED TO DELIVERY INSPECTION AND WHILE YOU DOING THIS INSPECTION JOB, PLEASE REFERRING TO THE REMANUFACTURED COMPONENTS - STANDARD CONFIGURATION.

✓ GOOD CONDITION   X : BAD CONDITION   ⊗ : NONE

SKETCH DRAWING OR PHOTOES

Upper view



Lower view

NO.	ITEM	IN	REMARK
	Differential		
	Model		
	Serial Number		
	Wo No.		
1	Case		
2	Cage		
3	Coupling		
4	Cage		
5	Bevel gear		
6	Nut		
7	Cap		
8	Bolt mounting bevel gear		
9	Difflock		
	Stand		
	Painting		
	Packing		

NOTE : PLEASE MARK " " FOLLOWING TO THE ACTUAL CONDITION

OIL LEVEL	EMPTY OR DRAINED	
	PROPER LEVEL	

Inspected By

Approved by

MEKANIK

GL / SPV

PT SAPTAINDRA SEJATI

DIFFERENTIAL

DISASSEMBLY

CHECKSHEET

MACHINE MODEL

HD785-7

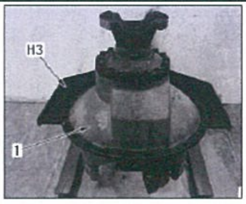
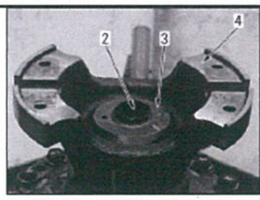
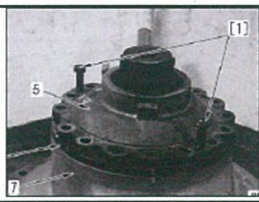
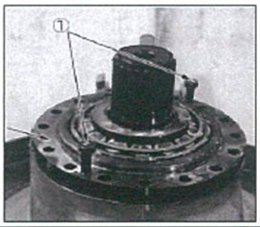
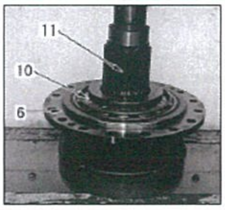
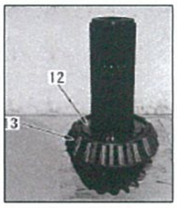
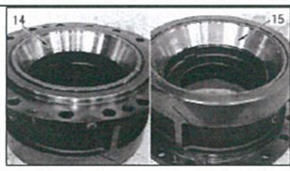
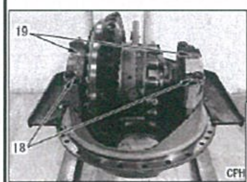
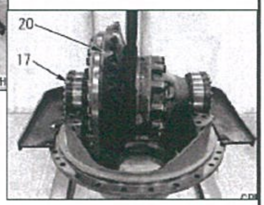
RO NO :

POWERTRAIN SECTION

PT.SAPTA INDRA SEJATI
PLANT REBUILD CENTER

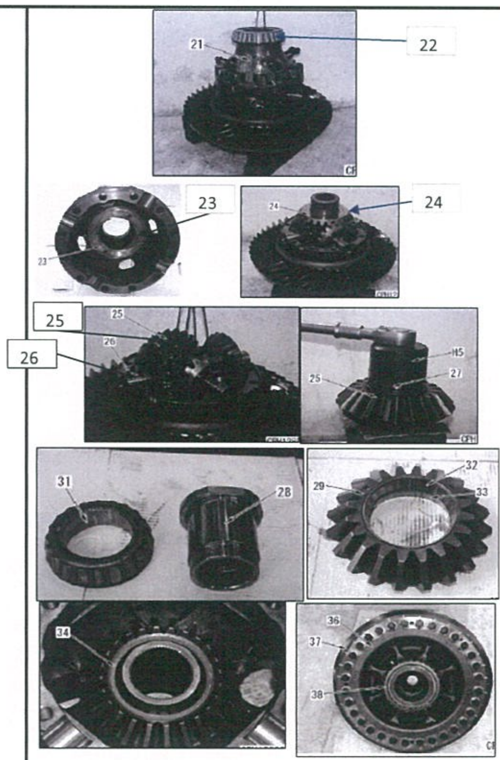
DISASSEMBLY CHECK SHEET **DIFFERENTIAL** HD 785 - 7				
	WO NUMBER		DOCUMENT NO	RC/QR02/PT/03/01
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
	UNIT MODEL		PAGE	

PROCESS	START		DISASSEMBLY BY		
	FINISH				

NO	PART TO REMOVE OR INSPECT	CONDITIONS TOBE INSPECTION	MECH	GL	SKECTH DRAWING OR PHOTOES	REMARK
1	DIFFERENTIAL ASSY	• Place differential assembly (1) on STAND				
2	COUPLING	• Remove mounting bolt (2), and remove coupling **(4) together with retainer (3) and O-ring.**				
3	CAGE ASSY	CONDITIONS TOBE INSPECTION • Screw in forcing screws [1], and remove cover (5). *Check the number and thickness of shims.* • Remove dust seal (8) from cover (5).				
4	PINION AND CAGE ASSY	• Screw in forcing screws [1], and lift pinion and cage assembly (9). • Lift and remove pinion and cage assembly (9).				
		• Remove cage (6) from pinion gear (11) together with bearing inner (10). • Remove spacer (12) and remove bearing inner (13).			 	
		• Remove bearing from cage. • Remove outer races (14) and (15).				
5	CAP ASSY	• Remove lock. • Loosen side bearing adjustment nuts. on both sides until it can be loosened by hand using special tool. • Loosen mounting bolts (18) and remove caps (19). • Lift and remove differential gear assembly (20). • Remove side bearing adjustment nuts (17) on both sides.			 	

PT.SAPTA INDRA SEJATI
PLANT REBUILD CENTER

DISASSEMBLY CHECK SHEET **DIFFERENTIAL** HD 785 - 7	WO NUMBER		DOCUMENT NO	RC/QR02/PT/03/01
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	
	UNIT MODEL		PAGE	

6	DIFFERENTIAL GEAR ASSEMBLY	• Remove mounting bolt and remove case • Remove bearing (22). • Remove thrust washer (23). • Remove side gear (24). • Remove pinion gear assembly (25) and cross shaft (26) together. • Fix pinion gear assembly (25) with a press, and remove ring nut (27) using special tool(h5). • Pull out shaft with a press, and remove bearing from pinion gear . • Remove bearing(31) from shaft(28) . • Remove bearing outer races and from pinion gear. • Remove side gear (34). • Remove thrust washer. • Remove bevel gear (37) from case (36). • Remove bearing (38) from the case.				

NOTE :

INSPECTION BY

APPROVED BY

MECHANIC

GL /QA

PT SAPTAINDRA SEJATI

DIFFERENTIAL MEASUREMENT CHECKSHEET

MACHINE MODEL

HD785-7

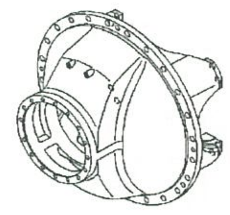
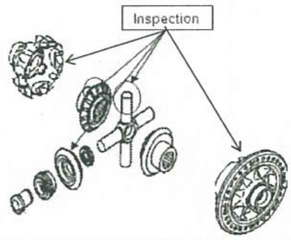
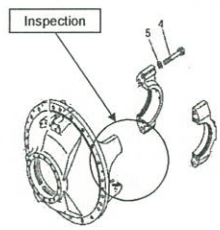
RO NO :

POWERTRAIN SECTION

PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

DISASSEMBLY CHECK SHEET **DIFFERENTIAL** **HD785-7**	WO NUMBER		DOCUMENT NO	
	COMP. MODEL		PUBLIC.DATE	08/09/2010
	COMP. S/N		REVISION NO.	00
	UNIT MODEL		PAGE	1 of 2

PROCESS	START		DISASSEMBLY BY	
	FINISH			

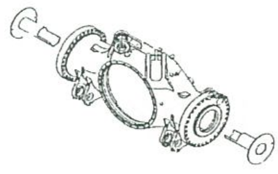
No	PARTS NAME	METHODE	CHECK POINT	EVALUATION ACTUAL	DECISION U/A	U/R	R/N	FIGURE OF PARTS TO BE CHECKED
1	CASE ASS'Y	VISUAL CHECK	Check Crack or Damage	RH / LH: X-X, Y-Y, Z-Z; CASE 1 / CASE 2: wear; wear	RH / LH			
2	COUPLING	VISUAL CHECK MEASUREMENT	Wear at oil seal contact surface DEPTH : STD ≤ 0. 1 mm WIDTH : STD ≤ 0. 5 mm Mating Face Contact With universal Joint ***Maximum wear less than 0,20 mm***	SEAL CONTACT (DEPTH, WIDTH) / MATING FACE: Wear				
3	CASE	VISUAL CHECK	Seat Seal contact, Thread hole condition, Condition from Wear and Crack or Damage					
4	CROSS SHAFT AND CASE	VISUAL CHECK MEASUREMENT	Measure and check dimension at contact area with sleeve pinion. ***Maximum wear less than 0,20 mm***	SHAFT / GEAR / PINION: Wear, Crack, Pitting	Shaft / Gear / Pinion			
5	SIDE GEAR	VISUAL CHECK	External gear teeth (Crack, Pitting, Breakage, Abnormal wear condition), Spline Wear condition ***Maximum wear less than 0,35 mm***					
6	PINION GEAR	VISUAL CHECK	External gear teeth (Crack, Pitting, Breakage, Abnormal wear condition)					
7	BEVEL GEAR	VISUAL CHECK MEASUREMENT	External gear teeth (Crack, Pitting, Breakage, Abnormal wear condition)	CONDITION / PERCENTAGE: Pitting, Crack, Scoring				
8	CASE AND CAP	VISUAL CHECK MEASUREMENT	Dimension at Bearing Seat Diameter, Thread hole condition	RH / LH: X-X, Y-Y, Z-Z	RH / LH			
9	PINION	VISUAL CHECK MEASUREMENT	External gear teeth (Crack, Pitting, Breakage, Abnormal wear condition)					

MODEL UNIT	DIMENSION STD (mm)
WA500	
WA600	
WA800	
HD 465-5	
HD785-5/7	Ø279,984 ~ Ø280,016

PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

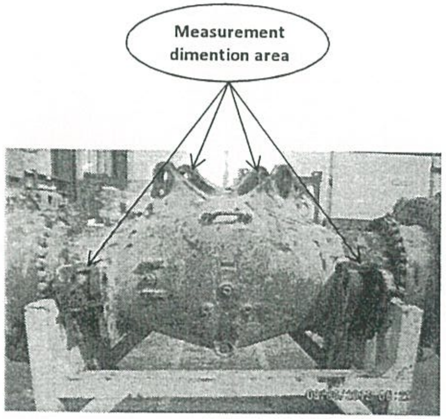
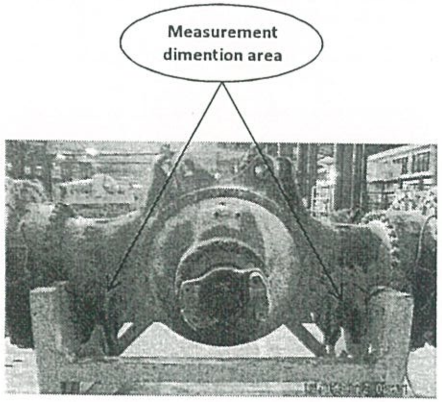
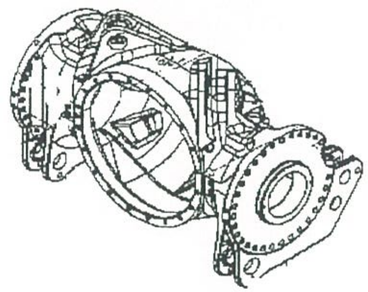
DISASSEMBLY CHECK SHEET	WO NUMBER		DOCUMENT NO	
DIFFERENTIAL	COMP. MODEL		PUBLIC.DATE	08/09/2010
HD785-7	COMP. S/N		REVISION NO.	00
	UNIT MODEL		PAGE	1 of 2

PROCESS	START		DISASSEMBLY BY	
	FINISH			

No	PARTS NAME	METHODE	CHECK POINT	EVALUATION ACTUAL	DECISION U/A	DECISION U/R	DECISION R/N	FIGURE OF PARTS TO BE CHECKED
1	HOUSING ASS'Y	VISUAL CHECK	Check Crack or Damage,check hole area contact bushing torque rod, measure dimention of inner diameter see,Drawing dimension	UPPER (RH / LH): X-X, Y-Y, Z-Z LOWER (RH / LH): X-X, Y-Y, Z-Z				

INSPECTION & MEASURING CHECKSHEET HOUSING BEFORE SALVAGE	PT SAPTAINDRA SEJATI NAROGONG REBUILD CENTER					
	Unit model	HD 785-7	Measuring date		Mechanic	QA Assurance
	Unit code		Time Start		1	
	Comp.Name		Time Finish			
	Comp.S/N				2	
	Cust.name					

U/A	Use Again
U/R	Use after Repair
R/N	Replace with New

GAMBAR	Qt'y	Kondisi yang harus diperiksa
 	1	1. Periksa kondisi Housing pada area Hole dudukan bushing 2. measure dimention of inner diameter yang yang contact dengan bushing Standard size : 104.000 -104.023 mm Roudness : 0.02 mm Cylinder city : 0.02 mm HOUSING P/N 561-22-73114 

Hasil pungukuran

Actual Pengukuran

X1	A	B	C	D	E	F	G	H
	I	J	K	L				
X2	A	B	C	D	E	F	G	H
	I	J	K	L				

Actual Pengukuran

Y1	A	B	C	D	E	F	G	H
	I	J	K	L				
Y2	A	B	C	D	E	F	G	H
	I	J	K	L				

PT SAPTAINDRA SEJATI

DIFFERENTIAL ASSEMBLY CHECKSHEET

MACHINE MODEL

HD785-7

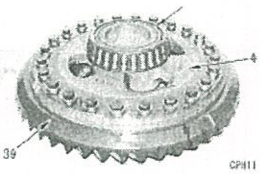
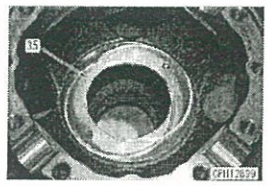
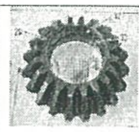
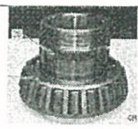
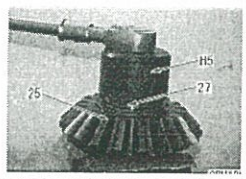
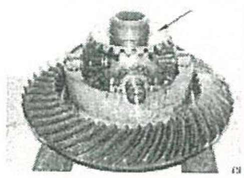
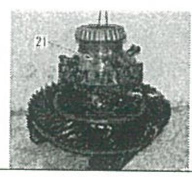
RO NO :

POWERTRAIN SECTION

PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

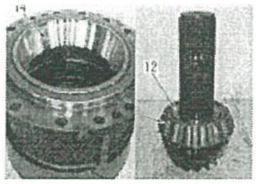
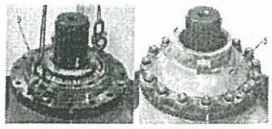
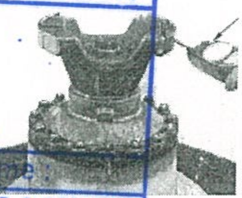
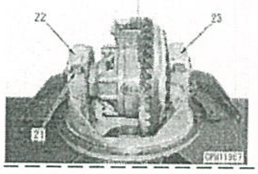
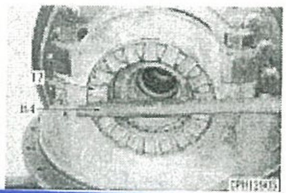
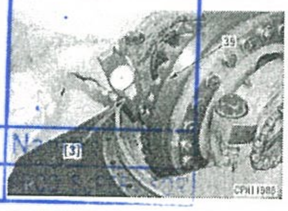
ASSEMBLY CHECK SHEET **DIFFERENTIAL** **HD785-7**	MACHINE MODEL		DOC.NO.	
	COMP. MODEL		PUB. DATE	01/08/2009
	COMP.S/N		REV. NO	00
	WO.NO		PAGE	1 of 3

PROCESS	START		ASSEMBLY BY	
	FINISH			

NO	PARTS	CHECK POINT	SPECIFICATION	SIGN MEC	SIGN SPV	SIGN QA	FIGURE OF PARTS TO BE CHECKED
1	DIFF.GEAR CASE ASSY	- Install bevel gear to diff case Then torque bolt Act. : kgm Install bearing to case	- Std torque kgm 94 – 105 LT -2 - Shrinking fit 100 C up Make sure no clearance The bearing to surface				
		- Install Thrust washer to case and side gear Make sure the head of the Dowel pin is lower than The washer surface Act. : kgm	- Std 0.5 – 0.7 mm				
2	SIDE GEAR	- install outer race bearing To side gear - Install bearing to shaft	- Press fit bearing				 
		- Install Nut Then torque nut Act. : kgm	- Std torque 59 – 76 Kgm				
		- Install side gear to cross Shaft and case	- make sure the pinion Gear rotates smoothly				
	SIDE GEAR ASSY	- Install side bearing - Install case Then torque bolt Act. : kgm	- Std torque 84 – 105 Kgm - LT - 2				

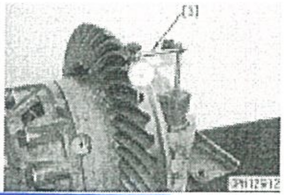
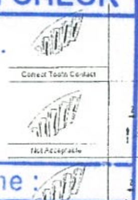
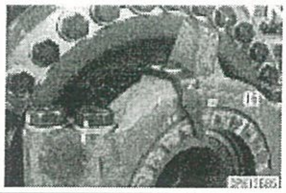
PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

ASSEMBLY CHECK SHEET **DIFFERENTIAL** **HD785-7**	MACHINE MODEL		DOC.NO.	
	COMP. MODEL		PUB. DATE	01/08/2009
	COMP.S/N		REV. NO	00
	WO.NO		PAGE	2 of 3

NO	PARTS	CHECK POINT	SPECIFICATION	MEC	SPV	QA	
3	PINION AND CAGE ASSY	- Install outer race to case - Install Bearing to shaft Make sure there is no Clearance outer race,inner Race to case and shaft	- Outer race press fit - Inner race bearing Shrinking fit 100 C				
		Install of pinion and cage - Use shims with same Number and thickness At disassembly Torque bolt Act. : kgm	- Std torque 50 – 62 Kgm - LT - 2 - Apply engine oil SAE 30 To the roller of the Bearing				
		- install coupling,Adj preload Rotate the bearing 20 -30 Before and after torque Monting bolt Act. : kgm Act. : kgm	- Std torque bolt 200 – 250 Kgm - LT - 2 - Std starting torque 5,1 Kgm or less	RANDOM ITEM CHECK Name : QA INSPECTOR			Name : PROD. SUPERVISOR 
4	DIFF GEAR ASSY	- install diff, gear assy to case. Torque bolt cap Act. : kgm	- Std torque 155 – 195 kgm - LT - 2				
		- adjustment Bearing preload make sure no clearence between ball bearing with outer race bearing	- Rotate bevel gear 20-30 Turn before adjusment				
5	ADJUSMENT PRELOAD	Preload of bearing. Act. : kgm	- Std preload 2.4 - 4.8 kgm - adjustment Nut for standard value	RANDOM ITEM CHECK Name : QA INSPECTOR			

PT SAPTAINDRA SEJATI
NAROGONG REBUILD CENTER

ASSEMBLY CHECK SHEET DIFFERENTIAL HD785-7	MACHINE MODEL		DOC.NO.	
	COMP. MODEL		PUB. DATE	01/08/2009
	COMP.S/N		REV. NO	00
	WO.NO		PAGE	3 of 3

NO	PARTS	CHECK POINT	SPECIFICATION	SIGN MEC	SIGN SPV	SIGN QA	FIGURE OF PARTS TO BE CHECKED
6	ADJUSTMENT BACK LASH	- Back Lash Act. : mm	- Std back lash 0.41 – 0.56 mm - measure at 3 – 4 places - When adjustment backlash/tooth contact, don't change preload of bearing.				
7	ADJUSTMEN TOOTH CONTACT	- Tooth contact Act. : %	- Std tooth contact 50 ~ 80 % - Adjust tooth contact at 7 or 8 teeth of driven gear.	RANDOM ITEM CHECK Name : QA INSPECTOR			Name : PROD. SUPERVISOR 
8	LOCK PLATE	- Install lock plate Torque monting bolt Act. :	- Std torque 11.5 ± 1 kgm - LT-2				

NOTE :

ASSEMBLY BY	INSPECTED BY	APPROVED BY
(MECHANIC)	(QA)	(GL / SPV)